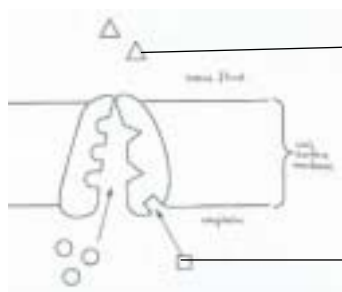


10(a)



K;

A;

2

Question	Answer	Marks
10(b)	any four from: 1 sodium (ion) channels open or Na⁺ enters (axon) ; 2 (so) membrane <u>depolarised</u> ; 3 <i>ref. to</i> threshold (potential) reached (for action potential to occur) ; 4 potassium (ion) channels open or K⁺ moves out (of axon) ; 5 (causes) <u>repolarisation</u> (of membrane) ; 6 <i>ref. to</i> <u>hyperpolarisation</u> (of membrane) ; 7 return to resting potential ; 8 AVP ; e.g. entry of sodium ions causes more sodium channels to open	4
10(c)	any four from: <i>description</i> 1 myelinated axon / myelination, produces faster transmission speed (than unmyelinated) ; 2 a larger the diameter (for unmyelinated) results in a faster transmission speed ; ora 3 myelination has a greater effect on transmission speed than axon diameter ; <i>explanation</i> 4 (myelinated faster due to) saltatory conduction or action potential jumps from node (of Ranvier) to node ; 5 axons with wider diameter have, less ion leakage / larger surface area / lower resistance ;	4